SafeLIBERO, Level I: one scene, two runs

"pick up the black bowl on the stove and place it on the plate"

obstacle: wine bottle next to the bowl's path
frozen VLA alone vs. with AG3S, same initial state and seed

Below, per axis: closest prior practice vs. AG3S
prior: earlier baseline runs on other LIBERO scenes
ours: the AG3S episode shown above

selected target surface
seeds / contact links
protected neighbor
attached object

VLA only



VLA + AG3S



VLA + AG3S, later



1 What: identity
the noun's attention names the object

Prior: segmenter



AG3S (ours)



2 Where it ends: extent
depth continuity sets the boundary

Prior: ellipsoid

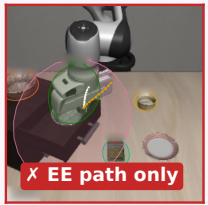


AG3S (ours)



3 Who and when: permission
contact links only, in the allowed stage

Prior: EE filter



AG3S (ours)

